Experiment No. 7

Title: Implementation of StyleGANs for generating high-quality images

Aim: To implement a Style-based Generative Adversarial Network (StyleGAN) for generating high-quality and diverse human face images using the CelebA dataset, and to demonstrate explicit control over the image generation process through a learned intermediate latent space, including coarse-to-fine style control, style mixing between two identities, and stochastic detail variation using noise injection.

Objectives:

1. To train the model on the CelebA face dataset for generating realistic human face images.
2. To understand the use of a mapping network for transforming the input latent vector from Z space into an intermediate W space.
3. To understand how style vectors are injected at different layers of the Generator to control coarse, middle-level, and fine image features.
4. To demonstrate style mixing by combining different style vectors from two latent identities.
5. To demonstrate noise injection for controlling stochastic details such as hair texture, skin details, and other fine variations.

Pre-Requisites:

1. Basic knowledge of Python programming.
2. Basic understanding of Deep Learning, CNNs and PyTorch.
3. Basic understanding of Generative Adversarial Networks (GANs), Generator and Discriminator.
4. Basic understanding of probability, latent vectors and adversarial optimization.
5. Basic familiarity with the CelebA dataset and image preprocessing.

Theory:

Style-Based Generative Adversarial Network

StyleGAN is a Generative Adversarial Network architecture designed to generate high-quality and highly controllable images, particularly human faces.

Unlike a conventional DCGAN, where a random latent vector z is directly supplied to the Generator, StyleGAN first transforms the latent vector into an intermediate latent representation called w.

The basic generation process can be represented as:

Random latent vector z → Mapping Network → Intermediate latent vector w → Synthesis Network → Generated Image

This separation provides greater control over different visual characteristics of the generated image.Tools and Technologies Required.

1. Google Colab
2. Python 3
3. PyTorch and Torchvision
4. NumPy
5. Matplotlib
6. CelebA dataset (via Kaggle)
7. GPU support in Google Colab (recommended)
8. Flask, HTML/CSS and JavaScript for the optional deployment extension

Procedure:

1. Open Google Colab and select a GPU runtime.
2. Import the required Python libraries such as PyTorch, Torchvision, NumPy, Matplotlib and PIL.
3. Download the CelebA dataset using the Kaggle API or use an available copy through Google Drive.
4. Extract the dataset and locate the aligned face images.
5. Display a few sample images from the CelebA dataset to understand the input data.
6. Define image preprocessing operations such as:
    - Resize the images.
    - Center crop the images if required.
    - Convert images to tensors.
    - Normalize pixel values.
7. Create a custom dataset class or use an appropriate image-folder dataset loader.
8. Create a DataLoader to provide batches of training images.

Conclusion:

The experiment successfully demonstrates the implementation of a Style-based Generative Adversarial Network (StyleGAN) for generating high-quality human face images using the CelebA dataset. Students understand how StyleGAN extends the conventional GAN/DCGAN

approach by introducing a mapping network, intermediate latent space, layer-wise style control and noise injection.